

EES Question Bank

1. What is the primary cause of global warming?

- A) Deforestation
- B) Greenhouse gas emissions
- C) Ocean currents
- D) Solar radiation

B) Greenhouse gas emissions

2. Which of the following gases is a major contributor to global warming?

- A) Oxygen
- B) Carbon dioxide
- C) Nitrogen
- D) Argon

B) Carbon dioxide

3. Which of the following human activities is most responsible for increasing carbon dioxide levels in the atmosphere?

- A) Agricultural activities
- B) Industrial processes
- C) Fossil fuel combustion
- D) Solar energy production

C) Fossil fuel combustion

4. What is the effect of global warming on sea levels?

- A) Sea levels decrease
- B) Sea levels remain the same
- C) Sea levels rise
- D) Sea levels fluctuate unpredictably

C) Sea levels rise

5. Which of the following is NOT a renewable energy source that can help reduce global warming?

- A) Wind energy
- B) Solar energy
- C) Natural gas
- D) Hydropower

C) Natural gas (It is a fossil fuel)

6. What is the greenhouse effect?

- A) A natural process where the Earth's atmosphere traps heat from the sun
- B) A phenomenon where the ozone layer blocks sunlight
- C) The cooling of the Earth due to cloud cover
- D) The warming of the Earth caused by volcanic eruptions

A) A natural process where the Earth's atmosphere traps heat from the sun

7. Which international agreement aims to limit global warming to below 2°C compared to pre-industrial levels?

- A) The Paris Agreement
- B) The Montreal Protocol
- C) The Rio Earth Summit
- D) The Geneva Convention

A) The Paris Agreement

8. What is acid rain?

- a) Rainwater with a pH greater than 7
- b) Rainwater with a pH less than 7
- c) Rainwater that has high levels of oxygen
- d) Rainwater that is neutral in pH

b) Rainwater with a pH less than 7 (Typically below 5.6)

9. Which of the following gases are primarily responsible for the formation of acid rain?

- a) Nitrogen dioxide and sulfur dioxide
- b) Oxygen and carbon dioxide
- c) Methane and ozone
- d) Nitrogen and hydrogen

a) Nitrogen dioxide and sulfur dioxide

10. Acid rain can damage buildings and monuments, particularly those made of which material?

- a) Glass
- b) Marble and limestone
- c) Wood
- d) Steel

b) Marble and limestone (Due to calcium carbonate reacting with acid)

11. What is the main cause of ozone depletion?

- a) Air pollution
- b) CFCs (Chlorofluorocarbons)
- c) Carbon dioxide
- d) Nitrogen oxides

b) CFCs (Chlorofluorocarbons)

12. Which layer of the Earth's atmosphere contains the ozone layer?

- a) Thermosphere
- b) Stratosphere
- c) Mesosphere
- d) Troposphere

b) Stratosphere

13. What is the effect of ozone depletion on humans?

- a) Increases plant growth
- b) Reduces the incidence of skin cancer
- c) Increases the risk of skin cancer
- d) Improves air quality

c) Increases the risk of skin cancer

14.What happens when ozone in the stratosphere is depleted?

- a) Increased UV radiation reaches Earth
- b) Air quality improves
- c) Greenhouse gases are reduced
- d) Temperature of the Earth's surface decreases

a) Increased UV radiation reaches Earth

15.Which of the following is NOT a component of the environment?

- a) Air
- b) Water
- c) Solar energy
- d) Gravity

d) Gravity

16.The biotic components of the environment include:

- a) Air and water
- b) Plants and animals
- c) Soil and minerals
- d) Light and temperature

b) Plants and animals

17.Which of these is an example of an abiotic component?

- a) Human beings
- b) Plants
- c) Sunlight
- d) Animals

b) Plants and animals

18.The term 'biodiversity' refers to:

- a) The variety of life on Earth
- b) The extinction of species
- c) The impact of pollution on the environment
- d) The process of environmental conservation

a) The variety of life on Earth

19.Which of these is a major environmental issue caused by deforestation?

- a) Soil erosion
- b) Water scarcity
- c) Increased biodiversity
- d) Global cooling

a) Soil erosion

20.What does the "Refuse" principle in the 5R concept primarily encourage?

- A) Reusing materials
- B) Recycling products
- C) Rejecting items that are unnecessary or harmful
- D) Rotting organic waste

C) Rejecting items that are unnecessary or harmful

21.What is the main focus of the "Reuse" principle in the 5R concept?

- A) Recycling materials into new products
- B) Buying more products
- C) Using items more than once to extend their life
- D) Reducing carbon footprints

C) Using items more than once to extend their life

22.Which of the following is NOT part of the 5Rs concept?

- A) Refuse
- B) Recycle
- C) Restore
- D) Reuse

C) Restore

23.Which of the following is NOT a Sustainable Development Goal (SDG)?

- a) Quality Education
- b) Zero Hunger
- c) Economic Growth
- d) Peace and Justice

c) Economic Growth

24.How many Sustainable Development Goals (SDGs) are there in total?

- a) 10
- b) 12
- c) 17
- d) 20

c) 17

25.Which of the following is the goal of SDG 7?

- a) Affordable and Clean Energy
- b) Clean Water and Sanitation
- c) Climate Action
- d) Responsible Consumption and Production

a) Affordable and Clean Energy

26.What is the target year for achieving the Sustainable Development Goals (SDGs)?

- a) 2025
- b) 2030
- c) 2040
- d) 2050

b) 2030

27.What is the primary focus of SDG 13?

- a) Climate Action
- b) Responsible Consumption
- c) Clean Water and Sanitation
- d) Good Health and Well-being

a) Climate Action

28.Which SDG is concerned with "clean water and sanitation"?

- a) SDG 6
- b) SDG 7
- c) SDG 8
- d) SDG 11

a) SDG 6

29.Which of the following texts mentions the reverence for natural elements and emphasizes the protection of the environment?

- a) Vedas
- b) Upanishads
- c) Ramayana
- d) Mahabharata

a) Vedas

30.In Vedic times, which natural element was considered sacred and worshipped as a deity?

- a) Fire (Agni)
- b) Sun (Surya)
- c) Wind (Vayu)
- d) All of the above

d) All of the above

31.The Pre-Vedic period was characterized by which of the following practices to conserve natural resources?

- a) Agricultural development
- b) Sacred groves and natural sanctuaries
- c) Animal sacrifice
- d) Metal working

b) Sacred groves and natural sanctuaries

32.In Vedic culture, how was the river Ganga regarded?

- a) As a source of irrigation
- b) As a purifier and sacred river
- c) As a trade route
- d) As a source of spiritual teachings

b) As a purifier and sacred river

33.Which of the following is NOT a primary function of forests?

- A) Oxygen production
- B) Carbon dioxide absorption
- C) Soil erosion prevention
- D) Manufacturing of agricultural equipment

D) Manufacturing of agricultural equipment

34.What is overexploitation of natural resources?

- A) The sustainable use of natural resources
- B) The excessive use of natural resources beyond their capacity to regenerate
- C) The controlled use of natural resources
- D) The preservation of natural resource

B) The excessive use of natural resources beyond their capacity to regenerate

35.Which of the following is a major consequence of overexploitation of forests?

- A) Biodiversity conservation
- B) Soil erosion
- C) Increase in wildlife population
- D) Decrease in atmospheric CO₂

B) Soil erosion

36.What is the main component of a solar panel?

- a) Copper wire
- b) Glass
- c) Silicon
- d) Aluminium

c) Silicon

37.What is the main component of a wind turbine that captures wind energy?

- a) Blades
- b) Generator
- c) Tower
- d) Rotor

a) Blades

38.What is biomass energy?

- a) Energy derived from fossil fuels
- b) Energy obtained from organic materials
- c) Energy obtained from wind
- d) Energy derived from water

b) Energy obtained from organic materials

39.Which of the following is NOT a type of ocean energy?

- A) Tidal Energy
- B) Wave Energy
- C) Geothermal Energy
- D) Ocean Thermal Energy

C) Geothermal Energy

40.Which technology is used to capture energy from the temperature difference between warm surface water and cold deep water?

- A) Tidal turbines
- B) Ocean thermal energy conversion (OTEC)
- C) Wave energy converters
- D) Solar panels

B) Ocean thermal energy conversion (OTEC)

41.Which of the following is a disadvantage of tidal energy?

- A) It is unpredictable
- B) It requires high water velocity
- C) It can impact marine ecosystems
- D) It cannot be used in coastal regions

C) It can impact marine ecosystems

42.Which of the following is the primary source of hydrogen?

- A) Water
- B) Coal
- C) Petroleum
- D) Uranium

A) Water

43.In which part of the Earth is geothermal energy primarily stored?

- a) Crust
- b) Mantle
- c) Core
- d) Atmosphere

b) Mantle

44.Geothermal energy is considered renewable because:

- a) It can be replenished by the Earth's heat
- b) It does not release harmful emissions
- c) It is inexhaustible
- d) All of the above

d) All of the above

45. Ecosystem includes:

- A) Only plants
- B) Only animals
- C) Living and non-living components
- D) Only microorganisms

C) Living and non-living components

46.Producers in an ecosystem are:

- A) Animals
- B) Plants
- C) Decomposers
- D) Parasites

B) Plants

47.Which is NOT a function of ecosystem?

- A) Energy flow
- B) Nutrient cycling
- C) Decomposition
- D) Industrial growth

D) Industrial growth

48. Genetic diversity refers to variation within:

- A) Ecosystem
- B) Species
- C) Habitat
- D) Community

B) Species

49.Which is a major cause of biodiversity loss?

- A) Conservation
- B) Afforestation
- C) Deforestation
- D) Sustainable use

C) Deforestation

50.Loss of biodiversity leads to:

- A) Stability
- B) Ecosystem balance
- C) Species extinction
- D) Sustainability

C) Species extinction

51.Which organization conducts biodiversity assessment in India?

- A) WHO
- B) UNEP
- C) Botanical Survey of India
- D) NASA

C) Botanical Survey of India

52.SWOT analysis of biodiversity hotspot helps in:

- A) Industrial planning
- B) Environmental assessment
- C) Urban expansion
- D) Agricultural mechanization

B) Environmental assessment

53.“Strength” in SWOT analysis of biodiversity hotspot refers to:

- A) Threats
- B) Weaknesses
- C) Natural advantages
- D) Pollution

C) Natural advantages

54.Which of the following is a major threat to biodiversity hotspots in India?

- A) Afforestation
- B) Wildlife protection
- C) Deforestation
- D) Conservation laws

C) Deforestation

55.In-situ conservation means:

- A) Conservation in laboratory
- B) Conservation in natural habitat
- C) Conservation in zoos
- D) Conservation in seed banks

B) Conservation in natural habitat

56.Which act is related to biodiversity conservation in India?

- A) Forest Rights Act
- B) Wildlife Protection Act, 1972
- C) Motor Vehicles Act
- D) IT Act

B) Wildlife Protection Act, 1972

57.National parks and wildlife sanctuaries are examples of:

- A) Ex-situ conservation
- B) In-situ conservation
- C) Gene banks
- D) Botanical gardens

B) In-situ conservation

58.The Wildlife Protection Act in India was enacted in:

- A) 1950
- B) 1962
- C) 1972
- D) 1986

C) 1972

59.Which act provides legal framework for biodiversity conservation in India?

- A) Environment Protection Act, 1986
- B) Biological Diversity Act, 2002
- C) Forest Rights Act, 2006
- D) Air Act, 1981

B) Biological Diversity Act, 2002

60.Which international agreement focuses on conservation of biodiversity?

- A) Kyoto Protocol
- B) Montreal Protocol
- C) Convention on Biological Diversity
- D) Paris Agreement

C) Convention on Biological Diversity

61.The Biological Diversity Act, 2002 mainly aims to:

- A) Promote mining
- B) Regulate access to biological resources
- C) Increase urbanization
- D) Promote deforestation

B) Regulate access to biological resources

62. Which organism produces food in an ecosystem?

- A) Consumers
- B) Decomposers
- C) Producers
- D) Parasites

C) Producers

63.Nutrient cycling in an ecosystem refers to:

- A) Flow of energy
- B) Recycling of matter
- C) Loss of nutrients
- D) Heat transfer

B) Recycling of matter

64. Decomposers play an important role in:

- A) Food production
- B) Energy generation
- C) Breaking down dead matter
- D) Photosynthesis

C) Breaking down dead matter

65. The flow of energy in an ecosystem is:

- A) Cyclic in nature and repeatedly reused
- B) Random and uncontrolled
- C) Unidirectional from producers to consumers
- D) Same at all trophic levels

C) Unidirectional from producers to consumers

66. One important characteristic of an ecosystem is interdependence, which means:

- A) Organisms live independently without interaction
- B) Only plants depend on animals
- C) All organisms depend on each other for survival
- D) Only decomposers are dependent on others

C) All organisms depend on each other for survival

67. The function of an ecosystem that helps in maintaining soil fertility is:

- A) Energy flow through food chains
- B) Regulation of population
- C) Nutrient cycling through decomposers
- D) Loss of heat energy

C) Nutrient cycling through decomposers

68. The main role of food chains in an ecosystem is to:

- A) Show recycling of matter
- B) Demonstrate energy transfer between organisms
- C) Explain climate regulation
- D) Increase ecosystem productivity

B) Demonstrate energy transfer between organisms

69. Loss of one species in an ecosystem can:

- A) Improve ecosystem functioning
- B) Have no impact on ecosystem
- C) Disturb the ecological balance
- D) Increase biodiversity

C) Disturb the ecological balance

70. The primary source of energy for most ecosystems is:

- A) Heat from the earth
- B) Energy stored in soil
- C) Solar energy from the sun
- D) Chemical energy from water

C) Solar energy from the sun